

EE450 Socket Programming — Grading Criteria

Spring 2026 • Total: 100 pts • Deductions applied on top of phase scores

Phase A1: Boot Up [40 pts total — 10 pts per server]

- hospital_server boots correctly — 10 pts
 - UDP port = 25000+xxx | TCP port = 26000+xxx
 - "Hospital Server is up and running using UDP on port <port>."
- authentication_server boots correctly — 10 pts
 - UDP port = 21000+xxx
 - "Authentication Server is up and running using UDP on port <port>."
- appointment_server boots correctly — 10 pts
 - UDP port = 23000+xxx
 - "Appointment Server is up and running using UDP on port <port>."
- prescription_server boots correctly — 10 pts
 - UDP port = 22000+xxx
 - "Prescription Server is up and running using UDP on port <port>."

Phase 1B: Authentication [15 pts total — 5 pts per case]

Log in successful — Patient [5 pts]

- Client terminal
 - "The client is up and running."
 - "<username> sent an authentication request to the hospital server."
 - "<username> received the authentication result. Authentication successful. You have been granted patient access."
- Hospital Server terminal
 - "Hospital Server received an authentication request from a user with hash suffix <hash_suffix>."
 - "Hospital Server has sent an authentication request to the Authentication Server."
 - "Hospital server has received the response from the authentication server using UDP over port <hospital UDP port>."
 - "User with a hash suffix <hash_suffix> has been granted access to the system. Determining the access of the user."
 - "User with hash <hash_suffix> will be granted patient access."
 - "Hospital Server has sent the response from Authentication Server to the client using TCP over port <hospital TCP port>."
- Authentication Server terminal

- "Authentication Server has received an authentication request for a user with hash suffix: <hash_suffix>."
- "Authentication succeeded for a user with hash suffix: <hash_suffix>."
- "The Authentication Server has sent the authentication result to the Hospital Server."

Log in successful — Doctor [5 pts]

- Client terminal
 - "The client is up and running."
 - "<username> sent an authentication request to the hospital server."
 - "<username> received the authentication result. Authentication successful. You have been granted doctor access."
- Hospital Server terminal
 - "Hospital Server received an authentication request from a user with hash suffix <hash_suffix>."
 - "Hospital Server has sent an authentication request to the Authentication Server."
 - "Hospital server has received the response from the authentication server using UDP over port <hospital UDP port>."
 - "User with a hash suffix <hash_suffix> has been granted access to the system. Determining the access of the user."
 - "User with hash suffix <hash_suffix> will be granted doctor access."
 - "Hospital Server has sent the response from Authentication Server to the client using TCP over port <hospital TCP port>."
- Authentication Server terminal
 - "Authentication Server has received an authentication request for a user with hash suffix: <hash_suffix>."
 - "Authentication succeeded for a user with hash suffix: <hash_suffix>."
 - "The Authentication Server has sent the authentication result to the Hospital Server."

Fail to log in [5 pts]

- Client terminal
 - "<username> sent an authentication request to the hospital server."
 - "The credentials are incorrect. Please try again."
- Hospital Server terminal
 - "Hospital Server received an authentication request from a user with hash suffix <hash_suffix>."
 - "Hospital Server has sent an authentication request to the Authentication Server."
 - "Hospital server has received the response from the authentication server using UDP over port <hospital UDP port>."

- "Hospital Server has sent the response from Authentication Server to the client using TCP over port <hospital TCP port>."
- Authentication Server terminal
 - "Authentication Server has received an authentication request for a user with hash suffix: <hash_suffix>."
 - "Authentication failed for a user with hash suffix: <hash_suffix>."
 - "The Authentication Server has sent the authentication result to the Hospital Server."

Phase 2: lookup [3 pts]

Correctly display the doctors on file [3 pts]

- Client terminal
 - "<patient_username> sent a lookup request to the hospital server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "The following doctors are available:"
 - "<doctor1_username>"
 - "<doctor2_username>"
 - "..."
- Hospital Server terminal
 - "Hospital Server received a lookup request from a user with a hash suffix <hash_suffix> over port <hospital TCP port>."
 - "Hospital Server sent the doctor lookup request to the Appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over port <hospital UDP port>."
 - "Hospital Server has sent the doctor lookup to the client."
- Appointment Server terminal
 - "The Appointment Server has received a doctor availability request."
 - "The Appointment Server has sent the lookup result to the Hospital Server."

Phase 2: lookup <doctor> [9 pts total — 3 pts per case]

If no appointment has been made (all slots available) [3 pts]

- Client terminal
 - "Patient <patient_username> sent a lookup request to the hospital server for <doctor_username>."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "All time blocks are available for <doctor_username>."

- Hospital Server terminal
 - "Hospital Server has received a lookup request from a user with hash suffix <hash_suffix> to lookup <doctor_username> availability using TCP over port <hospital TCP port>."
 - "Hospital Server sent the doctor lookup request to the Appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over port <hospital UDP port>."
 - "The Hospital Server has sent the response to the client."
- Appointment Server terminal
 - "The Appointment Server has received a doctor availability request."
 - "All time blocks are available for <doctor_username>."
 - "The Appointment Server has sent the lookup result to the Hospital Server."

If the doctor appointment is full [3 pts]

- Client terminal
 - "Patient <patient_username> sent a lookup request to the hospital server for <doctor_username>."
 - "The client received the response from the Hospital Server using TCP over port <client port>."
 - "<doctor_username> has no time slots available."
- Hospital Server terminal
 - "Hospital Server has received a lookup request from a user with hash suffix <hash_suffix> to lookup <doctor_username> availability using TCP over port <hospital TCP port>."
 - "Hospital Server sent the doctor lookup request to the Appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over port <hospital UDP port>."
 - "The Hospital Server has sent the response to the client."
- Appointment Server terminal
 - "The Appointment Server has received a doctor availability request."
 - "<doctor_username> has no time slots available."
 - "The Appointment Server has sent the lookup result to the Hospital Server."

Else display all available slots (some available) [3 pts]

- Client terminal
 - "Patient <patient_username> sent a lookup request to the hospital server for <doctor_username>."
 - "The client received the response from the Hospital Server using TCP over port <client port>."
 - "<doctor_name> is available at times:"
 - "<time block 1>"
 - "<time block 2>"
 - "..."

- Hospital Server terminal
 - "Hospital Server has received a lookup request from a user with hash suffix <hash_suffix> to lookup <doctor_username> availability using TCP over port <hospital TCP port>."
 - "Hospital Server sent the doctor lookup request to the Appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over port <hospital UDP port>."
 - "The Hospital Server has sent the response to the client."
- Appointment Server terminal
 - "The Appointment Server has received a doctor availability request."
 - "<doctor_username> has some time slots available."
 - "The Appointment Server has sent the lookup result to the Hospital Server."

Phase 2: schedule <doctor> <block period> <illness> [12 pts total — 3 pts per case]

Success [3 pts]

- Client terminal
 - "<patient_username> sent an appointment schedule request to the hospital server."
 - "The client received the response from the Hospital Server using TCP over port <client port>."
 - "An appointment has been successfully scheduled for patient <patient_username> with <doctor_username> at <start_time>."
- Hospital Server terminal
 - "Hospital Server has received a schedule request from a user with hash suffix: <hash_suffix> to book an appointment using TCP over port <hospital TCP port>."
 - "Hospital Server has sent the schedule request to the appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over <hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment scheduling request received (time: <time_block>, doctor: <doctor_name>, patient hash suffix: <hash_suffix>, illness: <illness>)."
 - "Appointment has been scheduled successfully for user <hash_suffix> with <doctor_username>."

Failure — Test outside of the hours [3 pts]

- Client terminal
 - "<patient_username> sent an appointment schedule request to the hospital server."

- "The client received the response from the hospital server using TCP over port <client port>."
- "Unable to schedule an appointment with <doctor_username> at <selected time>. Other available time blocks are"
- "<time_slot 1>"
- "<time_slot 2>"
- "..."
- Hospital Server terminal
 - "Hospital Server has received a schedule request from a user with hash suffix: <hash_suffix> to book an appointment using TCP over port <hospital TCP port>."
 - "Hospital Server has sent the schedule request to the appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over <hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment scheduling request received (time: <time_block>, doctor: <doctor_name>, patient hash suffix: <hash_suffix>, illness: <illness>)."
 - "The requested appointment time is not available."

Failure — Time block is already occupied [3 pts]

- Client terminal
 - "<patient_username> sent an appointment schedule request to the hospital server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "Unable to schedule an appointment with <doctor_username> at <selected time>. Other available time blocks are"
 - "<time_slot 1>"
 - "<time_slot 2>"
 - "..."
- Hospital Server terminal
 - "Hospital Server has received a schedule request from a user with hash suffix: <hash_suffix> to book an appointment using TCP over port <hospital TCP port>."
 - "Hospital Server has sent the schedule request to the appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over <hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment scheduling request received (time: <time_block>, doctor: <doctor_name>, patient hash suffix: <hash_suffix>, illness: <illness>)."
 - "The requested appointment time is not available."

Failure — All time blocks taken [3 pts]

- Client terminal
 - "<patient_username> sent an appointment schedule request to the hospital server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "Unable to schedule an appointment with <doctor_username> at this time, as all time blocks have been taken up."
- Hospital Server terminal
 - "Hospital Server has received a schedule request from a user with hash suffix: <hash_suffix> to book an appointment using TCP over port <hospital TCP port>."
 - "Hospital Server has sent the schedule request to the appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over <hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment scheduling request received (time: <time_block>, doctor: <doctor_name>, patient hash suffix: <hash_suffix>, illness: <illness>)."
 - "The requested appointment time is not available."

Phase 2: view_appointment (patient) [6 pts total — 3 pts per case]

Appointment was found [3 pts]

- Client terminal
 - "<patient_username> sent a request to view their appointment to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "You have an appointment scheduled with <doctor name> at <time_slot>."
- Hospital Server terminal
 - "Hospital server has received a view appointment request from a user with hash suffix <hash_suffix> to view their appointment details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the view appointments request to the Appointment Server."
 - "Hospital Server has received the response from the appointment server using UDP over port <Hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal

- "Appointment Server has received a view appointment command for the user with hash suffix <hash_suffix>."
- "Returning details regarding the appointment for the user with hash suffix <hash_suffix>."

No appointment was found [3 pts]

- Client terminal
 - "<patient_username> sent a request to view their appointment to the Hospital Server."
 - "The client received the response from the hospital server using client port <client port>."
 - "You do not have an appointment today."
- Hospital Server terminal
 - "Hospital server has received a view appointment request from a user with hash suffix <hash_suffix> to view their appointment details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the view appointments request to the Appointment Server."
 - "Hospital Server has received the response from the appointment server using UDP over port <Hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment Server has received a view appointment command for the user with hash suffix <hash_suffix>."
 - "The user with hash suffix <hash_suffix> has no appointment in the system."

Phase 2: view_appointments (doctor) [6 pts total — 3 pts per case]

There is one or more appointments [3 pts]

- Client terminal (doctor)
 - "<doctor_username> sent a request to view their scheduled appointments to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "<doctor_name> is scheduled at times:"
 - "<time block 1>"
 - "<time block 2>"
 - "..."
- Hospital Server terminal

- "Hospital Server has received a view appointments request from <doctor_username> to view their schedule details using TCP over port <hospital TCP port>."
- "The hospital server has sent the view appointments request to the Appointment Server."
- "Hospital server has received the response from the Appointment server using UDP over port <Hospital UDP port>."
- "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment Server has received a request to view appointments scheduled for <doctor_username>."
 - "Returning the scheduled appointments for <doctor_username>."

All time slots are available [3 pts]

- Client terminal (doctor)
 - "<doctor_username> sent a request to view their scheduled appointments to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "You do not have any appointments scheduled."
- Hospital Server terminal
 - "Hospital Server has received a view appointments request from <doctor_username> to view their schedule details using TCP over port <hospital TCP port>."
 - "The hospital server has sent the view appointments request to the Appointment Server."
 - "Hospital server has received the response from the Appointment server using UDP over port <Hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment Server has received a request to view appointments scheduled for <doctor_username>."
 - "No appointments have been made for <doctor_username>."

Phase 2: cancel [6 pts total — 3 pts per case]

Success [3 pts]

- Client terminal
 - "<patient_username> sent a cancellation request to the Hospital Server."
 - "The client received the response from the Hospital Server using TCP over port <client port>."
 - "You have successfully cancelled your appointment with <doctor_username> at <time_slot>."

- Hospital Server terminal
 - "Hospital Server has received a cancel request from user with hash suffix: <hash_suffix> to cancel their appointment using TCP over port <hospital TCP port>."
 - "The hospital server has sent the cancel request to the appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over port <hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment Server has received a cancel appointment command for the user with hash suffix: <hash_suffix>."
 - "Successfully cancelled appointment."
- Note: timeslot line must remain in appointments.txt; only patient hash and illness are removed

Failure — No appointment found [3 pts]

- Client terminal
 - "<patient_username> sent a cancellation request to the Hospital Server."
 - "The client received the response from the Hospital Server using TCP over port <client port>."
 - "You have no appointments available to cancel."
- Hospital Server terminal
 - "Hospital Server has received a cancel request from user with hash suffix: <hash_suffix> to cancel their appointment using TCP over port <hospital TCP port>."
 - "The hospital server has sent the cancel request to the appointment server."
 - "Hospital Server has received the response from Appointment Server using UDP over port <hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment Server has received a cancel appointment command for the user with hash suffix: <hash_suffix>."
 - "Error: Failed to find appointment."

Phase 3: prescribe <patient_username> <frequency> (doctor) [3 pts]

Success [3 pts]

- Client terminal (doctor)
 - "<doctor_username> sent a request to the Hospital Server to prescribe <patient_username> following their diagnosis."

- "The client received the response from the hospital server using TCP over port <client port>."
 - "You have successfully prescribed <patient_username> with <treatment>, to be taken <frequency>."
- Hospital Server terminal
 - "Hospital Server has received a prescription request from <doctor_username> for a user with hash suffix <hash_suffix> using TCP over port <hospital TCP port>."
 - "Hospital Server has sent a request to fetch patients with hash suffix <hash_suffix> illness information to the Appointment Server."
 - "Hospital Server has received the illness response from the Appointment server using UDP over port <Hospital UDP port>."
 - "Acquiring treatment for <illness> from the database."
 - "Hospital server has sent the prescription request to the prescription server to prescribe <treatment>."
 - "Hospital server has received the response from the prescription server using UDP over port <Hospital UDP port>."
 - "The hospital server has sent the response to the client."
- Appointment Server terminal
 - "Appointment Server has received a request from Hospital Server regarding information about a user with hash suffix <hash_suffix> from <doctor_username>."
 - "Sending back the requested information to the Hospital server."
 - "Successfully removed <hash_suffix> appointment slot, <time_block> is now free to be scheduled for tomorrow."
- Prescription Server terminal
 - "Prescription Server has received a request from <doctor_username> to prescribe the user with hash suffix <hash_suffix>."
 - "Successfully saved the prescription details for user with hash suffix: <hash_suffix>."

Phase 3: view_prescription (patient) [9 pts total — 3 pts per case]

Prescription exists (active frequency) [3 pts]

- Client terminal (patient)
 - "<patient_username> sent a request to view their prescription to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "You have been prescribed <treatment>, to be taken <frequency>, by <doctor_username>."

- Hospital Server terminal
 - "Hospital Server has received a prescription request from a patient with hash suffix <hash_suffix> to view their prescription details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the prescription request to the Prescription Server."
 - "Hospital server has received the response from the prescription server using UDP over port <hospital UDP port>."
 - "Hospital server has sent the response to the client."
- Prescription Server terminal
 - "The prescription server has received a request to view the prescription for the user with hash suffix: <hash_suffix>."
 - "A prescription exists for this user."

Frequency value is None [3 pts]

- Client terminal (patient)
 - "<patient_username> sent a request to view their prescription to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "You were not prescribed any treatment by <doctor_username> following your diagnosis."
- Hospital Server terminal
 - "Hospital Server has received a prescription request from a patient with hash suffix <hash_suffix> to view their prescription details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the prescription request to the Prescription Server."
 - "Hospital server has received the response from the prescription server using UDP over port <hospital UDP port>."
 - "Hospital server has sent the response to the client."
- Prescription Server terminal
 - "The prescription server has received a request to view the prescription for the user with hash suffix: <hash_suffix>."
 - "There are no current prescriptions for this user."

No prescription record exists [3 pts]

- Client terminal (patient)
 - "<patient_username> sent a request to view their prescription to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "You do not have a prescription to look up."
- Hospital Server terminal

- "Hospital Server has received a prescription request from a patient with hash suffix <hash_suffix> to view their prescription details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the prescription request to the Prescription Server."
 - "Hospital server has received the response from the prescription server using UDP over port <hospital UDP port>."
 - "Hospital server has sent the response to the client."
- Prescription Server terminal
 - "The prescription server has received a request to view the prescription for the user with hash suffix: <hash_suffix>."
 - "There are no current prescriptions for this user."

Phase 3: view_prescription <patient_username> (doctor) [6 pts total — 3 pts per case]

Prescription exists [3 pts]

- Client terminal (doctor)
 - "<doctor_username> sent a request to view <patient_username> prescription to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."
 - "<patient_username> has been prescribed <treatment>, to be taken <frequency>, by <doctor_username_from_entry>."
- Hospital Server terminal
 - "Hospital Server has received a prescription request from <doctor_username> to view a patient with hash suffix <hash_suffix> prescription details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the prescription request to the Prescription Server."
 - "Hospital server has received the response from the prescription server using UDP over port <hospital UDP port>."
 - "Hospital server has sent the response to the client."
- Prescription Server terminal
 - "The prescription server has received a request to view the prescription for the user with hash suffix: <hash_suffix>."
 - "A prescription exists for this user."

No prescription exists [3 pts]

- Client terminal (doctor)
 - "<doctor_username> sent a request to view <patient_username> prescription to the Hospital Server."
 - "The client received the response from the hospital server using TCP over port <client port>."

- "<patient_username> does not have a prescription."
- Hospital Server terminal
 - "Hospital Server has received a prescription request from <doctor_username> to view a patient with hash suffix <hash_suffix> prescription details using TCP over port <Hospital TCP port>."
 - "Hospital Server has sent the prescription request to the Prescription Server."
 - "Hospital server has received the response from the prescription server using UDP over port <hospital UDP port>."
 - "Hospital server has sent the response to the client."
- Prescription Server terminal
 - "The prescription server has received a request to view the prescription for the user with hash suffix: <hash_suffix>."
 - "There are no current prescriptions for this user."

Subtract Points

- -1: The username is displayed outside of the client
 - Only the last 5 hex characters of the hash (hash_suffix) should appear on any server terminal
 - Plaintext username must only appear inside the client terminal
- 10pts: Multiple clients cannot connect to the server simultaneously
 - 5pts if they attempted to create a parallel program, but it just didn't work
- 3pts each: Wrong or hardcoded static port numbers
 - Static ports must match 21000+xxx, 22000+xxx, 23000+xxx, 25000+xxx, 26000+xxx
 - Dynamic client ports must use getsockname()
- 1pts: Extra on-screen debug messages not commented out before submission
- 3pts: cancel removes the timeslot line from appointments.txt
 - Timeslot entry must remain; only hashed patient identifier and illness should be stripped
- 1pts: prescribe does not free the appointment slot
 - Patient hash and illness must be removed from the entry, leaving only the time
- 5/100 max: Submission does not compile (make all fails)
- 10/100 max: Compiles but crashes immediately with no tasks performed
- 0: Missing README or Makefile — submission will not be graded

Point Summary

- Phase A1 — Boot Up: 40 pts
- Phase 1B — Authentication: 15 pts (3 cases × 5)
- Phase 2 — lookup: 3 pts
- Phase 2 — lookup <doctor>: 9 pts (3 cases × 3)

- Phase 2 — schedule: 12 pts (4 cases × 3)
- Phase 2 — view_appointment (patient): 6 pts (2 cases × 3)
- Phase 2 — view_appointments (doctor): 6 pts (2 cases × 3)
- Phase 2 — cancel: 6 pts (2 cases × 3)
- Phase 3 — prescribe: 3 pts
- Phase 3 — view_prescription (patient): 9 pts (3 cases × 3)
- Phase 3 — view_prescription (doctor): 6 pts (2 cases × 3)
- -----
- TOTAL: 100 pts (before deductions)

Phase A1 — Boot Up: 40 pts

Phase 1B — Authentication: 10 pts (3 cases × 3)

Phase 2 — lookup: 2 pts

Phase 2 — lookup <doctor>: 8 pts (3 cases × 3)

Phase 2 — schedule: 10 pts (4 cases × 2)

Phase 2 — view_appointment (patient): 6 pts (2 cases × 3)

Phase 2 — view_appointments (doctor): 6 pts (2 cases × 3)

Phase 2 — cancel: 4 pts (2 cases × 2)

Phase 3 — prescribe: 4 pts

Phase 3 — view_prescription (patient): 6 pts (3 cases × 2)

Phase 3 — view_prescription (doctor): 4 pts (2 cases × 2)

TOTAL: 100 pts (before deductions)